Introducción a la Electrónica

Unidad 6

Fundamentos de la Realimentación Negativa

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Parte 3

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6.2.2. REALIMENTACIÓN DE TENSIÓN EN PARALELO
(sensa tensión y realimenta corriente)

Diagrama en bloques:

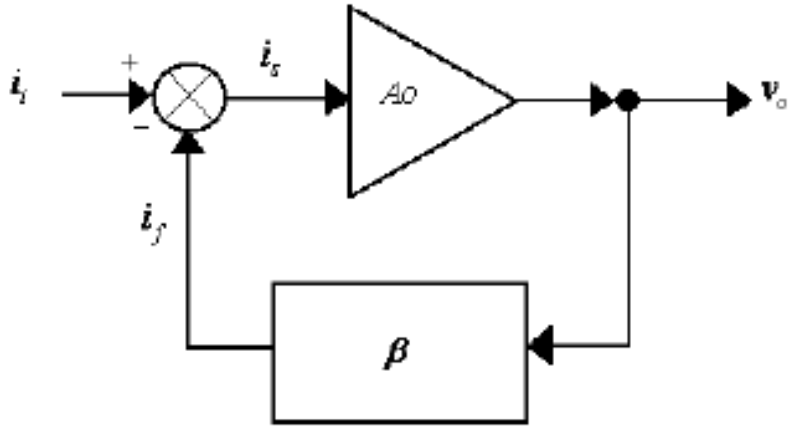


Fig. 6.5 – Diagrama en bloques correspondiente

$$Ao = \frac{v_o}{i_\varepsilon} \qquad (6.11)$$

$$\beta = \frac{i_f}{v_o} \qquad (6.12)$$

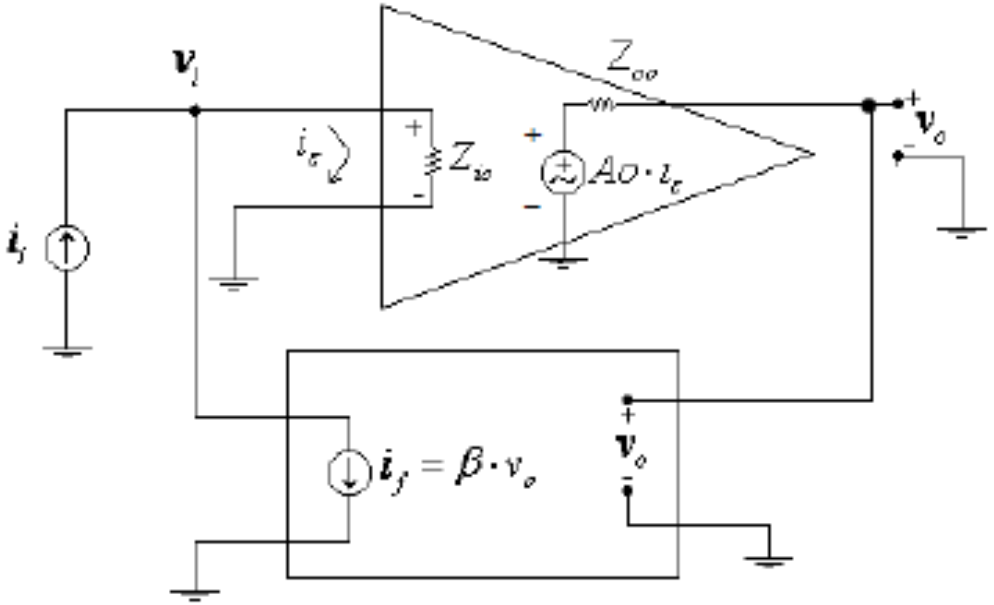


Fig. 6.6 – Esquema equivalente con cuadripolos

a) Transresistencia:

$$v_o = Ao \cdot i_s \left(Ao \equiv \text{Transresistencia} \right)$$

$$i_i = i_s + \beta \cdot v_o = i_s + Ao \cdot \beta \cdot i_s = i_s \left(1 + Ao \cdot \beta \right)$$

$$A_{RF} = \frac{v_o}{i_i} = \frac{Ao \cdot i_s}{i_s \left(1 + Ao \cdot \beta \right)} = \frac{Ao}{\left(1 + Ao \cdot \beta \right)} \qquad (6.13)$$

$$\left\{ \begin{matrix} Ao \equiv A_{oR}(\Omega) \\ \beta \equiv \beta_G (mhos) \end{matrix} \right\} \Rightarrow \left[A_{RF} \right] = \Omega$$

b) Ganancia de Tensión:

$$A_V = \frac{v_o}{v_i} = \frac{Ao \cdot i_s}{i_s \cdot Z_{io}} = \frac{Ao \cdot i_s}{i_s \cdot Z_{io}} = \frac{Ao}{Z_{io}} \left(\text{adimensional} \right) \qquad (6.14)$$

Impedancia de entrada

$$v_i = i_i \cdot Z_{iF} = i_\varepsilon \cdot Z_{io}$$

$$i_i = i_\varepsilon + i_f = i_\varepsilon \cdot (1 + \beta \cdot Ao)$$

$$i_i \cdot Z_{iF} = i_\varepsilon \cdot (1 + \beta \cdot Ao) \cdot Z_{iF} = i_\varepsilon \cdot Z_{io}$$

$$Z_{iF} = \frac{Z_{io}}{(1 + \beta \cdot Ao)}$$

Determinación de la impedancia de salida

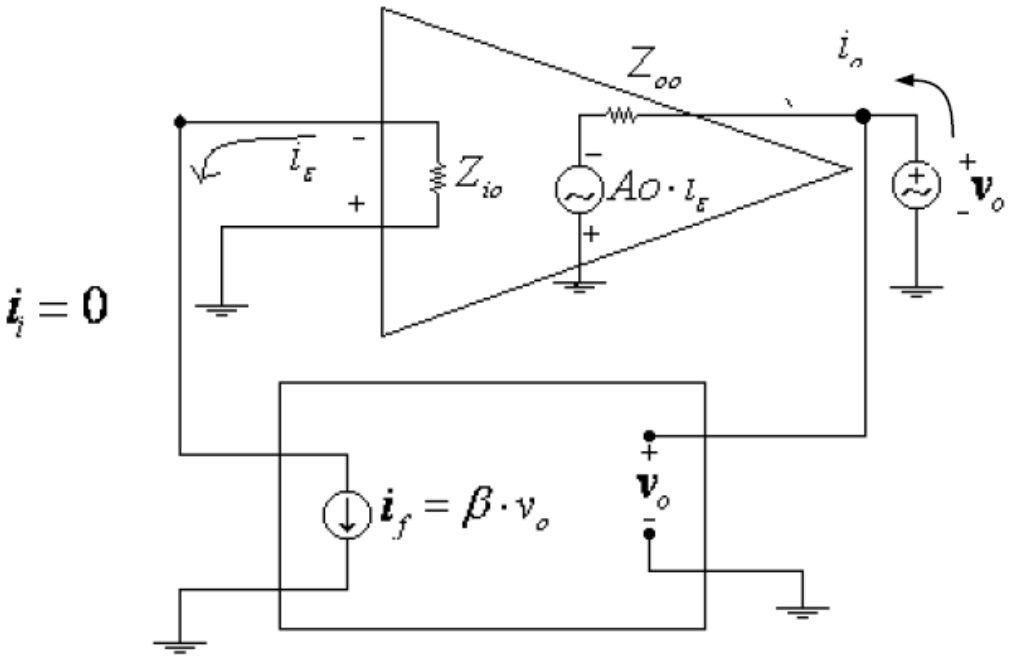


Fig. 6.7 – Esquema equivalente con cuadripolos

$$v_o = i_o \cdot Z_{oo} + Ao \cdot i_\varepsilon = i_o \cdot Z_{oo} - \beta \cdot Ao \cdot v_o \quad \Rightarrow$$

$$\Rightarrow \quad v_o \cdot (1 + \beta \cdot Ao) = i_o \cdot Z_{oo} \quad \Rightarrow$$

$$\Rightarrow \quad \frac{v_o}{i_o} = \frac{Z_{oo}}{(1 + \beta \cdot Ao)} = Z_{OF}$$

6.2.3. REALIMENTACIÓN DE CORRIENTE EN SERIE:
(sensa corriente y realimenta tensión)

Diagrama en bloques:

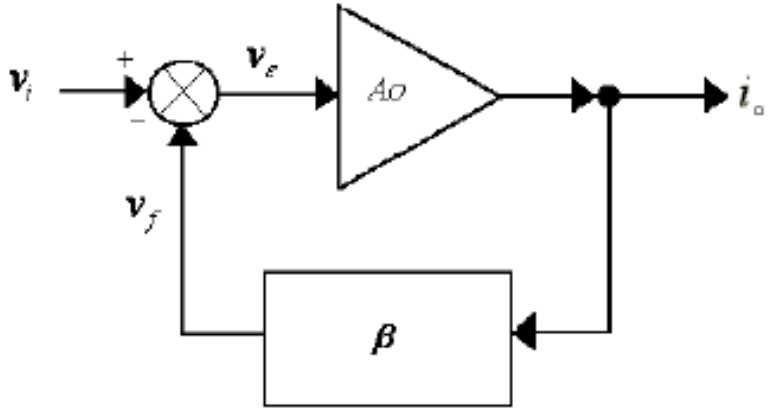


Fig. 6.8 – Diagrama en bloques correspondiente

$$A_O = \frac{i_O}{v_\varepsilon} \qquad (6.17)$$

$$\beta = \frac{v_f}{i_O} \qquad (6.18)$$

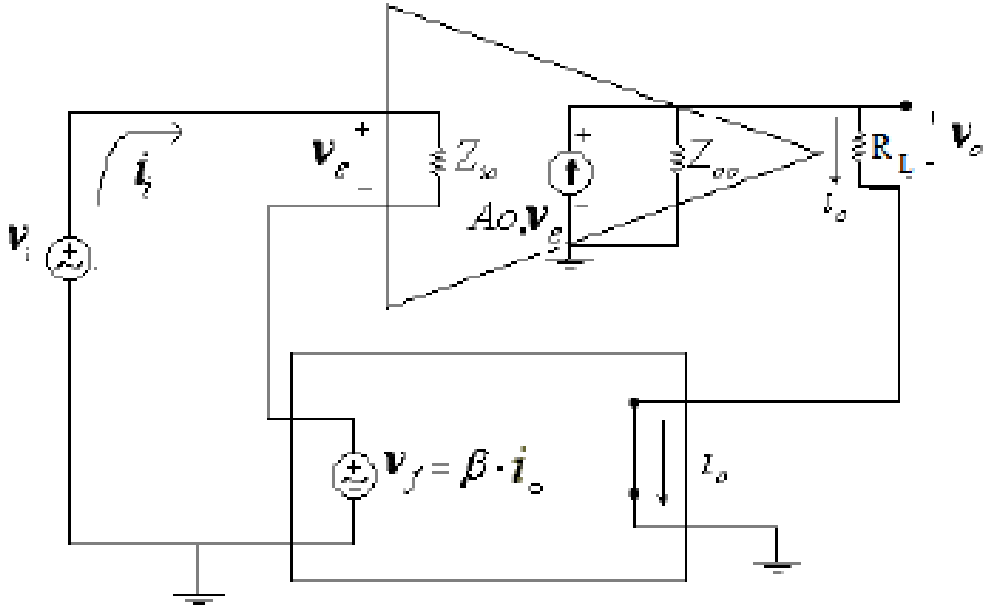


Fig. 6.9 – Esquema equivalente con cuadripolos

Transconductancia:

$$i_o = \frac{A_o . Z_{oo} . v_s}{Z_{oo} + R_L} = A_0' . v_s$$

$$v_i = v_s + \beta . i_0 = v_s + \beta . A_o' . v_s = v_s . (1 + \beta . A_o')$$

$$A_{GF} = \frac{i_o}{v_i} = \frac{A_o'}{1 + \beta . A_o'}$$

[Ao']=mhos, [β]=Ω

Ganancia de Tensión:

$$v_o = i_o . R_L$$

$$A_V = \frac{v_o}{v_i} = \frac{i_o . R_L}{v_i} = \frac{A_o' . R_L}{1 + A_o' \beta}$$

Ganancia de Corriente:

$$i_i = \frac{v_s}{Z_{io}}$$

$$i_o = A_o' . v_s$$

$$A_I = \frac{i_o}{i_i} = \frac{A_o' . v_s}{\frac{v_s}{Z_{io}}} = A_o' . Z_{io}$$

Impedancia de entrada

$$Z_{iF} = \frac{v_i}{i_i} = \frac{v_\varepsilon . (1 + \beta . A_o')}{\frac{v_\varepsilon}{Z_{io}}} = Z_{io} . (1 + \beta . A_o')$$

Impedancia de salida

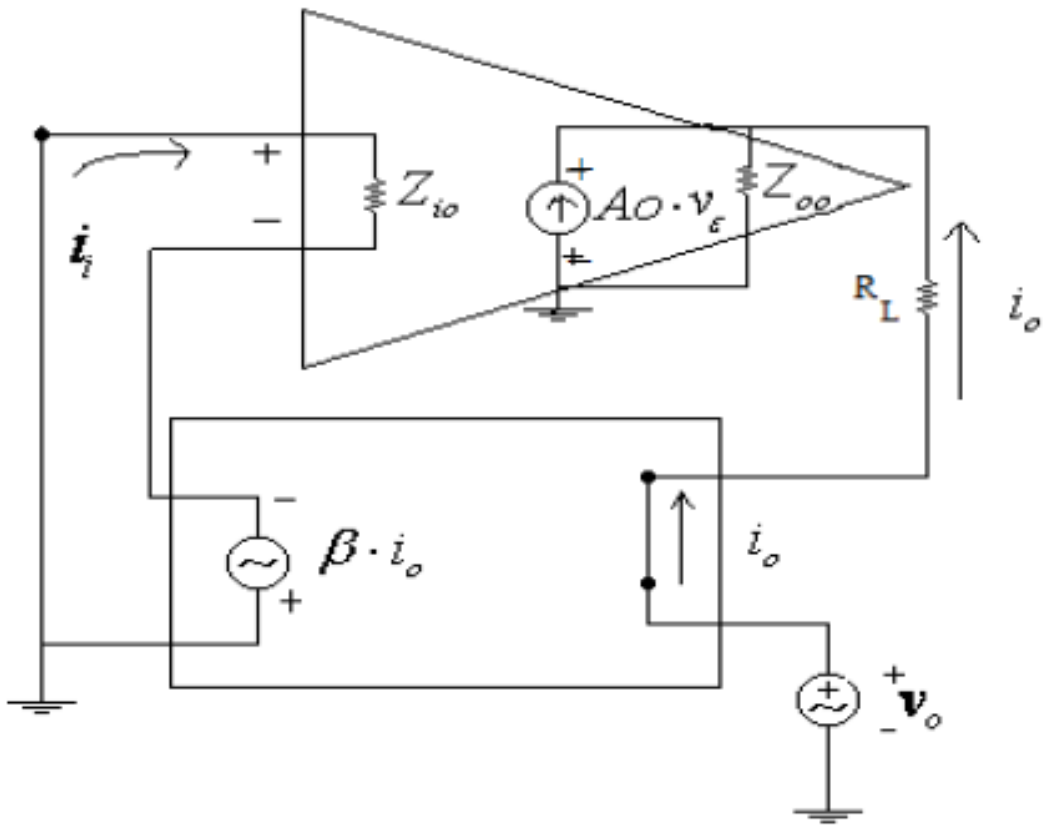
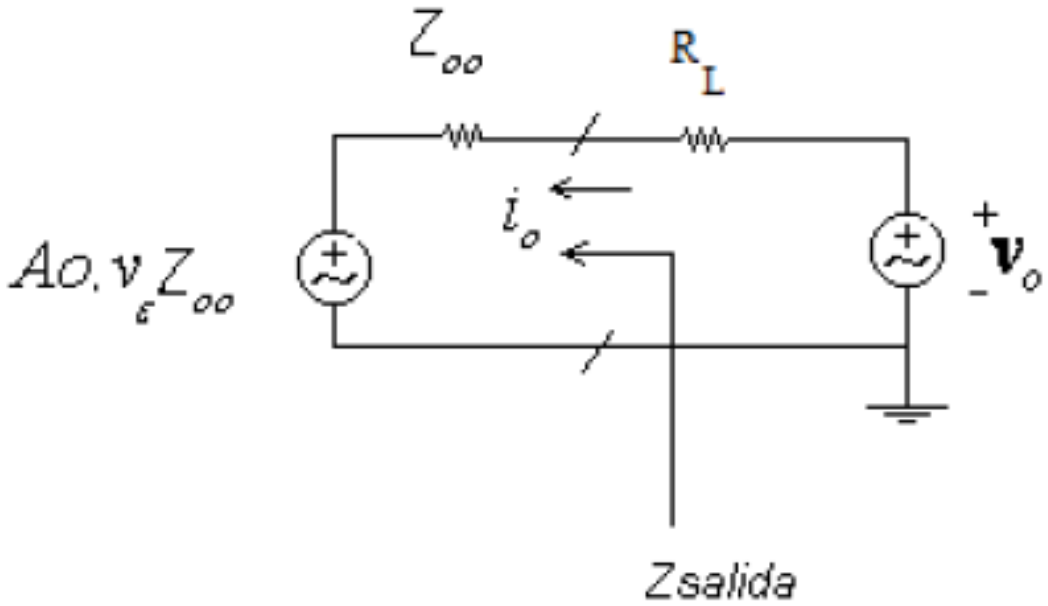


Fig. 6.10 – Esquema equivalente con cuadripolos

Impedancia de salida



$$v_o = i_o.R_L + i_o.Z_{oo} + A_o.v_\varepsilon.Z_{oo}$$

$$v_\varepsilon = \beta.i_o$$

$$v_o = i_o.R_L + i_o.Z_{oo} + i_o.\beta.A_o.Z_{oo}$$

$$v_o = i_o.[R_L + Z_{oo}.(1 + \beta.A_o)]$$

$$Z_{oF} = R_L + Z_{oo}.(1 + \beta.A_o)$$

Fig. 6.11 – Circuito equiv. para hallar $Z\ salida$

6.2.4. REALIMENTACIÓN DE CORRIENTE EN PARALELO
(sensa corriente y realimenta corriente)

Diagrama en bloques:

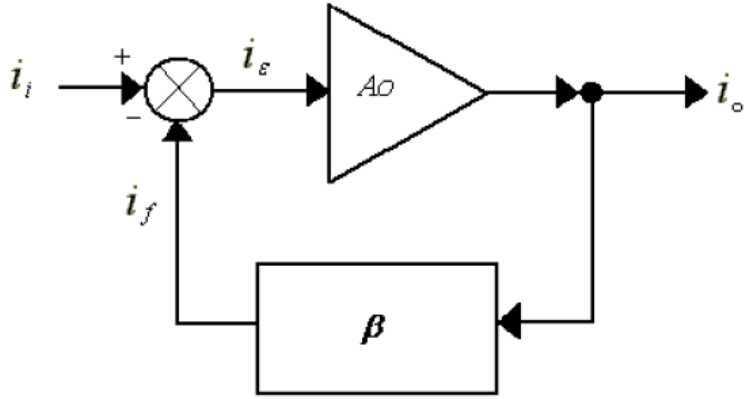


Fig. 6.12 – Diagrama en bloques correspondiente

$$A_O = \frac{i_O}{i_\varepsilon} \qquad (6.24)$$

$$\beta = \frac{i_f}{i_O} \qquad (6.25)$$

a) Ganancia de Corriente:

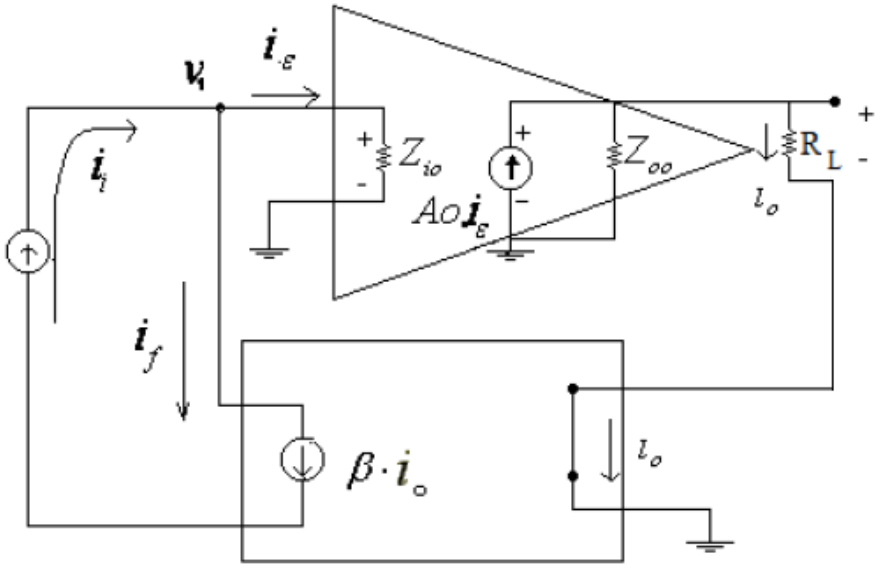


Fig. 6.13 – Esquema equivalente con cuadripolos

$$i_o = \frac{A_o . Z_{oo} . i_\varepsilon}{R_L + Z_{oo}} = A_o' . i_\varepsilon$$

$$i_i = i_\varepsilon + i_f = i_\varepsilon + \beta . A_o' . i_\varepsilon = i_\varepsilon . (1 + \beta . A_o')$$

$$A_{IF} = \frac{i_o}{i_i} = \frac{A_o'}{(1 + \beta . A_o')}$$

Ao, Ao' y β son adimensionales.

Ganancia de Tensión

$$v_o = i_o . R_L = A_o' . i_\varepsilon . R_L$$

$$v_i = v_I = i_\varepsilon . Z_{io}$$

$$A_V = \frac{v_o}{v_i} = \frac{A_o' . R_L}{Z_{io}}$$

Impedancia de entrada:

$$Z_{iF} = \frac{v_i}{i_i} = \frac{i_\varepsilon . Z_{io}}{i_\varepsilon . (1 + \beta . A_o')} = \frac{Z_{io}}{1 + \beta . A_o'}$$

Impedancia de salida

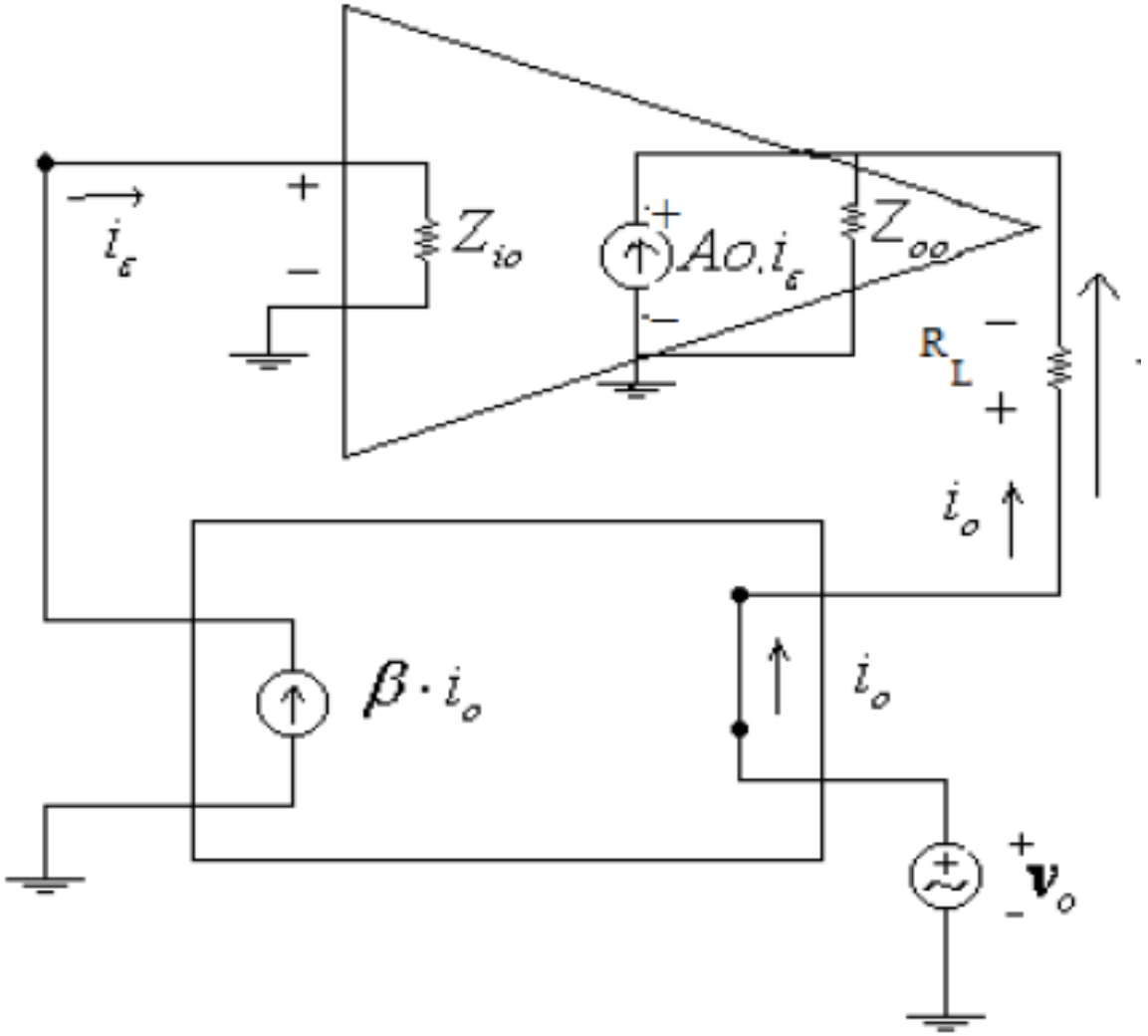


Fig. 6.13 – Esquema equivalente con cuadripolos

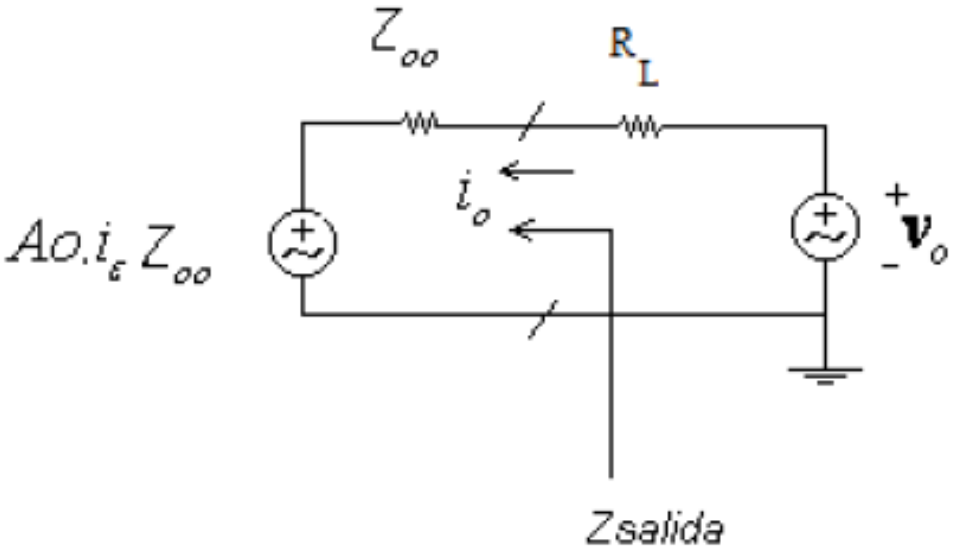


Fig. 6.14 – Circuito equiv. para hallar $Z\,salida$

$$v_o = i_o.R_L + i_o.Z_{oo} + A_o.i_\varepsilon.Z_{oo}$$

$$i_\varepsilon = \beta.i_o$$

$$v_o = i_o.R_L + i_o.Z_{oo} + i_o.\beta.A_o.Z_{oo}$$

$$v_o = i_o.[R_L + Z_{oo}.(1+\beta.A_o)]$$

$$Z_{oF} = R_L + Z_{oo}.(1+\beta.A_o) \qquad (6.29)$$